

**Configuration management:**

Externalising configuration and managing it centrally is essential for microservices. This approach helps manage different configurations for development, testing, and production environments, and simplifies the management of environment-specific settings.

**Resilience and fault tolerance:**

Microservices must be designed to be resilient to failures of other services or infrastructure components. Implementing patterns like circuit breaker, bulkhead, and retry can help manage partial failures without affecting the entire system.

**Continuous deployment/delivery:**

Microservices should be designed to be independently deployable via automated deployments. This approach supports frequent updates and improvements to individual microservices without impacting others.

**Security:** Each microservice can implement its own security, tailored to the specific needs and risks associated with its business functionality. Common practices include using OAuth for distributed security and ensuring sensitive data is encrypted both at rest and in transit.

**Monitoring and logging:**

Due to the distributed nature of microservices, centralised logging and monitoring are crucial. They provide insights into the health of individual services and the system as a whole, aiding in troubleshooting and performance tuning.

These design approaches provide the foundation for developing a robust, scalable, and maintainable microservices architecture, tailored to the specific needs and goals of a project or organisation.

## Popular microservices in cloud platforms

**Compute microservices:** These include AWS Lambda, Google Cloud Functions and Azure Functions, which are serverless compute services that

run code in response to events. Another such microservice is Kubernetes, a container orchestration platform that automates the deployment, scaling, and management of containers.

**Storage microservices:** Amazon S3, Google Cloud Storage, and Azure Blob Storage are object storage services that store and retrieve large amounts of data. Amazon Elastic Block Store (EBS) is a block-level storage service that provides persistent storage for EC2 instances.

**Database microservices:** Amazon RDS, Google Cloud SQL, and Azure SQL Database are relational database services that provide a managed database instance. Amazon DynamoDB and Google Cloud Firestore are NoSQL database services that provide a fast and flexible database. Azure Cosmos DB is a globally distributed, multi-model database service.

**Security microservices:**

AWS IAM is a security service that provides identity and access management for AWS resources. Google Cloud IAM is a security service that provides identity and access management for Google Cloud resources. Azure Active Directory (AAD) provides identity and access management for Azure resources, and AWS Cognito is a security service that provides user identity

and access management for web and mobile applications.

**Networking microservices:**

AWS VPC, Google Cloud Virtual Network, and Azure Virtual Network are virtual networking services that provide a secure and isolated network environment. AWS Route 53 is a domain name system (DNS) service that provides routing and traffic management.

**Analytics microservices:**

Amazon Redshift, Google Cloud BigQuery, Azure Synapse Analytics, and AWS Lake Formation are data warehousing services that provide fast and efficient data analysis.

**Machine learning microservices:**

AWS SageMaker, Google Cloud AI Platform and Azure Machine Learning are machine learning services that provide a platform for building, training, and deploying machine learning models.

**IoT microservices:** AWS IoT Core, Google Cloud IoT Core, and Azure IoT Hub are IoT services that provide a platform for connecting, monitoring, and managing IoT devices.

These are just a few examples of the many microservices available in cloud platforms. Each microservice provides a specific functionality that can be used to build, deploy, and manage applications in the cloud. **END** 🐧

## References

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